

Challenge — Titanic Survival Prediction

Your challenge

Now it's your turn!

You will train a machine learning model to answer a simple question:

Can we predict whether a passenger survived the Titanic disaster?

For this challenge, we simplified the original Titanic dataset to keep only **3 features**:

- **Sex** → passenger's sex
- **Pclass** → passenger's ticket class
- **Fare** → ticket price

And our target is:

- **Survived** → **0** = No, **1** = Yes

Your mission

Starting from the dataset:

[titanic_simple.csv](#)

1. **Load the dataset**
2. **Check for missing values**
3. **Prepare **X** and **Y****
4. **Encode the **Sex** column**
5. **Split the data into training and testing sets**

6. Train an `MLPClassifier`
7. Make predictions
8. Calculate the accuracy
9. Plot the loss curve

Hints

You have already seen **every step** in the Iris example.

Your code should follow the same pipeline:

Data → **Preparation** → **Split** → **Scaling** → **MLP** → **Training** → **Prediction** → **Evaluation**

The only new step is that `Sex` contains text (`male` / `female`), so you need to convert it into numerical values before training.

★ Bonus challenge

Try changing:

```
hidden_layer_sizes= (5,)
```

or:

```
learning_rate_init=0.1
```

and train the model again.

Does the accuracy change? Does the loss curve change? Why?

Goal

Don't worry about getting **100% accuracy**.

The goal is not to find the perfect model.

The goal is to **take what you learned about neural-network training and apply it to a new dataset**.

